Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think the most useful thing in the development of my project will be the ability to use dataclasses because there will be multiple variables for each car and being able to sort those accordingly will be very helpful and used alot

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think sorting the information will the the easiest just because its relatively simple and the hardest part will definitely be making it work in drafter because im very unsure still on how to make it work

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think what will take the most effort will likely either be the search option or the

separate cars pages and making it possible to switch between them

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

When it comes to distractions it will just be my phone and that's pretty simple to avoid, when I work on the assignment the resources i may need will likely be the llm or the office hours. I don't have any questions about the assignment at this moment.

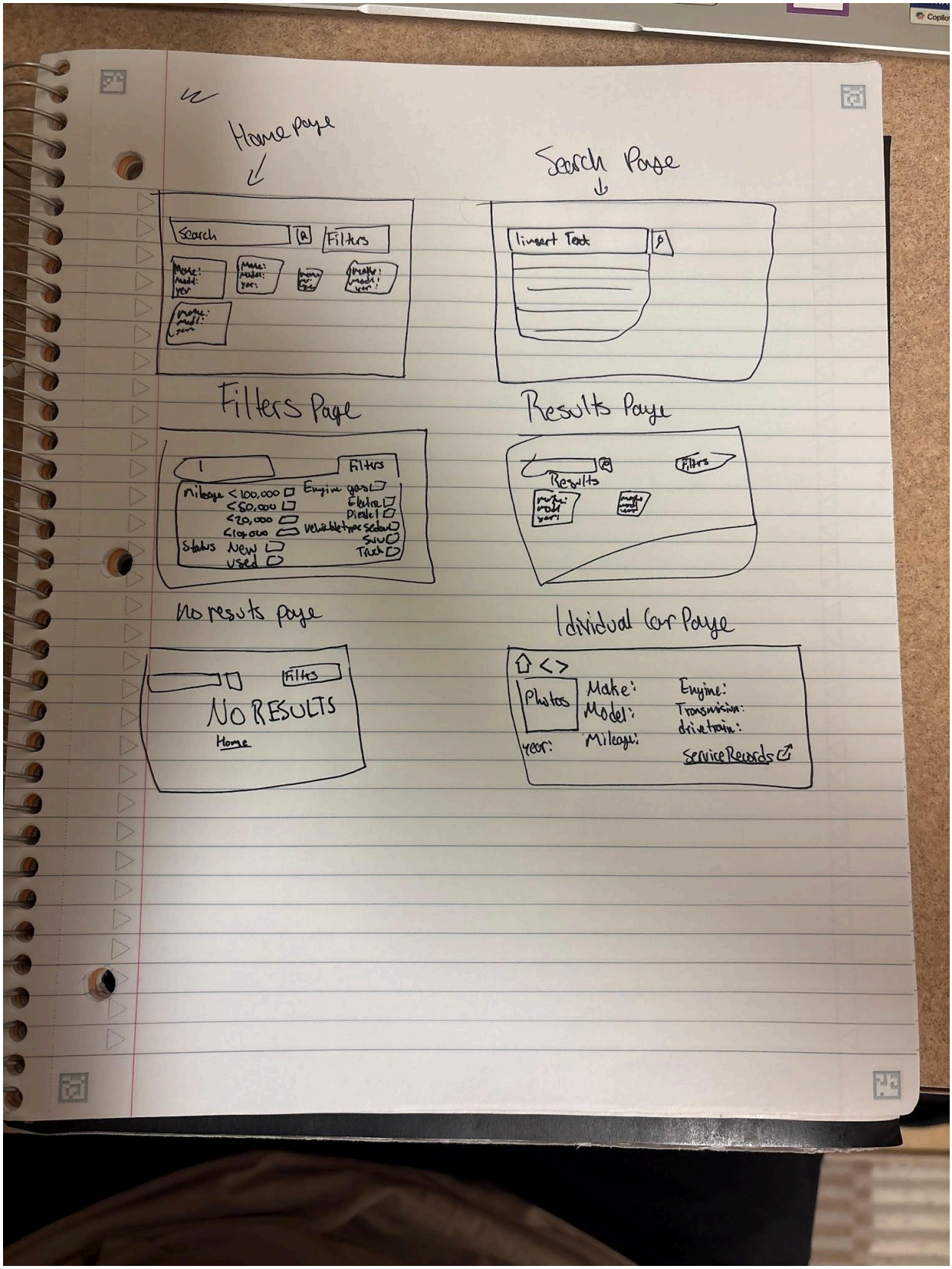
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Home page
Search
Filters
Search Page
Insert Text
Filters Page
Filters
Mileage
<100,000
<50,000
<20,000
<10,000
Status
New
Used
Engine
gas
Electric
Diesel
Vehicle type
Sedan
SUV
Truck
Results Page
Results
Filters
no results page
Filters
NO RESULTS
Home
Individual Car Page
Photos
Make:
Model:
Mileage:
Engine:
Transmission:
drivetrain:
year:
Service Records

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The purpose of the site is a place to shop for cars with different options and a way to search through the options and find the car that you're looking for.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	A datclass for each car showing what exactly it is named car Attributes: Year: int Make: str Model: str Mileage: int Service records: list[service] Service What done: str Date:str Location: str
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Service What done: str Date:str Location: str
Routes (minimum 7) The names of your route	I would have routes with names like, go_home, veiw_car, search, search_results, filter_options,back, no_results

functions	
Pages (minimum 5) A brief description of the pages in your site	There will be pages of the specific car that show the specific cars information, there will be multiple of these, there will be a home page that shows all of the options, there will be a search page that gives you places to enter information to find what you're searching for, there will be a search results page that shows the findings of your search and there will be a sort page to organize your search results
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	The impute that the user could search for would be some of the cars attributes like the make the model and the year, they may also be able to search for more generic things like the cars vehicle type all the things that people will search will be different variables that are held within the cars dataclasses all of the searches and filters will be strings or integers
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	There will be if statements when the user searches because there will be comparisons between the search inputs and the possible results, and for the filters there will be if statements to see if there are things that are the same as the things from the search or filter. And also if there are no results that match what the user imputed there will return a no_result page and if the if staemtns pass they will result in the results page with the cars in it that pass the if statement
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	I will likely use a few loops but i feel like the one that i wil use and get the most use of is searching through the dataclasses and finding the vehicles that match the search and adding them to a list of all the cars that will be displayed on the search results page

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I didnt use the llm at all in this assignment

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.